



Wireshark Display Filters

Use this when: you need the few packets that matter in a capture: one host, one conversation, errors or suspicious traffic.

Wireshark 4.x · Security+ · CCNA
Sheet v1.0 · Sep 2026

AUTHORIZED USE ONLY Capture only on networks you own or may monitor. Captures can hold passwords and personal data: store them encrypted and delete when done.

Anatomy of a display filter

Type it in the bar above the packet list

ip.addr	==	192.0.2.10	&&	tcp.port	in	{80,443}
FIELD protocol.field	OPERATOR	VALUE	LOGIC	FIELD	SET	VALUES

DISPLAY FILTER

CAPTURE FILTER

When	After capture, change anytime	Before capture, fixed
Syntax	Wireshark fields	BPF (libpcap)
Example	tcp.port == 80	tcp port 80
Use for	Finding packets	Smaller files

Green: valid filter

Red: syntax error

Yellow: check it

Operators

C-LIKE	ENGLISH	MEANING
==	eq	equal (any value matches)
!=	ne	not equal (all values, 3.6+)
> < >= <=	gt lt ge le	compare numbers
&& !	and or not	combine / negate
contains		field has this text
matches		regex, e.g. "(?i)login"
in {...}		in a set: {443, 8000..8080}

Everyday filters

ip.addr == 192.0.2.10	to or from a host
ip.src == 192.0.2.0/24	from a subnet
!(ip.addr == 192.0.2.10)	hide a host
tcp.port == 443 udp.port == 53	HTTPS or DNS
dns · dhcp · arp · icmp	one protocol
tcp.stream eq 5	one conversation
eth.addr == 00:00:5e:00:53:01	by MAC

Troubleshoot

tcp.analysis.flags	any TCP problem
tcp.analysis.retransmission	lost packets
tcp.flags.reset == 1	refused / reset
tcp.analysis.zero_window	receiver full
dns.flags.rcode != 0	DNS errors
http.response.code >= 400	web errors
arp.duplicate-address-detected	IP conflict
icmp.type == 3	unreachable

Security hunting

tcp.flags.syn == 1 && tcp.flags.ack == 0	SYN scans
ftp telnet http.authorization	cleartext logins
http.request.method == "POST"	form submits
tls.handshake.type == 1	Client Hello, see SNI
frame contains "password"	leaked secrets

CAUTION Don't run Wireshark as root/admin. Capture through `dumpcap` (the `wireshark` group on Linux) so the GUI stays unprivileged.



REMEMBER

- Right-click a field → **Apply as Filter**
- Capture filter `tcp port 80` ≠ display `tcp.port == 80`
- To exclude a host, use `!(ip.addr == x)`

- Text values need quotes: `"POST"`
- Follow** → **TCP Stream** rebuilds a conversation
- Save a filter as a button with the **+** on the bar